📊 STUDENT MENTAL HEALTH ANALYSIS USING EXPLORATORY DATA ANALYSIS (EDA)

🧠 1. INTRODUCTION

Mental health among students has become an important issue in recent years. Students face multiple challenges such as academic pressure, competition, financial stress, and lack of social interaction. These factors can significantly affect their mental well-being, leading to anxiety, depression, and stress.

This project focuses on analyzing student data using Exploratory Data Analysis (EDA) techniques to understand patterns, relationships, and key factors that influence mental health.

❗ 2. PROBLEM STATEMENT

Students experience various stress factors during their academic journey, including:

- 📚 High academic workload
- 💸 Financial concerns
- 🧍 Social isolation
- 🔮 Uncertainty about future careers

These issues can lead to serious mental health problems such as depression and anxiety.

👉 **Problem:**
There is a lack of data-driven understanding of which factors most strongly impact student mental health.

👉 **Goal:**
To analyze student data and identify the key factors affecting mental health using statistical and visualization techniques.

🎯 3. OBJECTIVES

The objectives of this project are:

- 📊 To understand the distribution of mental health-related variables
- 🔗 To analyze relationships between academic, social, and psychological factors
- 📈 To identify patterns and trends in the data
- 🚨 To detect outliers and unusual behavior

- 🔍 To determine the most important factors affecting depression
- 💡 To provide meaningful insights and recommendations

🛠️ 4. TOOLS AND TECHNOLOGIES USED

The following tools and technologies were used in this project:

- 🐍 Python
- 🧮 Pandas
- 🔢 NumPy
- 📉 Matplotlib
- 📊 Seaborn
- 💻 Jupyter Notebook
- ⚡ ydata-profiling (Auto EDA tool)

📂 5. DATASET DESCRIPTION

The dataset includes various features related to student life and mental health:

- 👤 Demographic features: Age, Gender
- 📚 Academic features: Academic workload, academic pressure
- 🤝 Social features: Social relationships
- 🏃 Lifestyle features: Sports engagement, sleep
- 🧠 Mental health indicators: Depression, anxiety, isolation
- 📌 Derived feature: Total score

These variables help analyze both academic and psychological aspects of students.

📊 6. EXPLORATORY DATA ANALYSIS (EDA)

EDA is performed to understand the structure, distribution, and relationships within the dataset.

📌 6.1 Statistical Analysis

The following statistical measures were calculated:

- 📍 Mean, Median, Mode
- 📉 Variance and Standard Deviation
- 🔄 Skewness and Kurtosis
- 📏 Range

👉 **Key observations:**

- Most variables are moderately distributed
- Some variables show slight skewness
- Mental health variables are not perfectly normally distributed

📌 6.2 Descriptive Summary

Using descriptive statistics:

- 📊 Count, mean, standard deviation, minimum, maximum, and quartiles were analyzed

👉 **Observations:**

- Age ranges between approximately 17 to 24
- Most mental health scores lie between 1 and 5
- Data distribution is fairly consistent

📌 6.3 Correlation Analysis

Correlation analysis was performed to identify relationships between variables.

👉 **Key findings:**

- 📈 Academic workload has a positive relationship with depression
- 🔥 Anxiety is strongly correlated with depression
- 🤝 Social relationships have a negative relationship with depression

This indicates that better social connections reduce mental stress.

📈 7. DATA VISUALIZATION

Visualization techniques were used to better understand patterns and relationships.

📊 7.1 Univariate Analysis

📌 **Histogram**
Used to understand the distribution of individual variables

📌 **KDE Plot**
Used to visualize probability density

📌 **Boxplot**
Used to detect outliers and spread

📌 **Count Plot**
Used to analyze frequency of categorical variables

📊 7.2 Bivariate Analysis

📌 **Scatter Plot**
Shows relationship between two variables
👉 Example: Academic workload vs Depression
👉 Insight: Higher workload leads to higher depression

📌 **Bar Plot**
Used to compare averages across categories

📌 **Line Plot**
Shows trends across academic years

📌 **Violin Plot**
Shows distribution + density
👉 Insight: Students involved in sports have lower anxiety

📌 **Boxplot (Feature vs Depression)**
Shows variation of depression across categories

📊 7.3 Multivariate Analysis

📌 **Pair Plot**
Shows relationships among multiple variables

📌 **Strip Plot**
Displays individual data points

📌 **Correlation Heatmap** 🔥
Shows strength of relationships

👉 **Key insight:**
Depression is strongly related to:

- Anxiety
- Academic pressure
- Isolation

🔍 8. KEY INSIGHTS

The analysis revealed the following important insights:

- 📈 Higher academic workload increases depression levels
- 😟 Anxiety and depression are strongly correlated
- 💸 Financial concerns contribute to mental stress
- 🧍 Isolation leads to higher depression
- 🤝 Strong social relationships reduce stress
- 🏃 Sports engagement helps reduce anxiety
- 🎓 Degree level shows slight variation in stress levels
- 📊 Total score is a strong indicator of overall mental health.

📌 9. FEATURE IMPORTANCE (FUTURE ML STEP)

Feature importance was analyzed using correlation with depression.

👉 **Most important features:**

- Anxiety
- Isolation
- Academic pressure
- Financial concerns

This can be used for building machine learning models.

👥 10. GROUP ANALYSIS

Group analysis was performed based on degree level.

👉 Compared:

- Depression
- Anxiety
- Academic pressure

👉 **Observation:**
Postgraduate students show slightly higher stress levels.

⚙️ 11. AUTO EDA

The ydata-profiling tool was used for automated analysis.

👉 **Benefits:**

- Fast data overview
- Missing value detection
- Correlation insights
- Summary reports

✅ 12. CONCLUSION

This project successfully applied EDA to understand student mental health.

👉 **Key conclusions:**

- Academic pressure is a major factor
- Anxiety and depression are closely linked
- Lifestyle and social factors play a key role

EDA helped uncover meaningful insights and patterns.

💡 13. RECOMMENDATIONS

🎓 For Students:

- Maintain work-life balance
- Engage in physical activities
- Build strong social connections

🏫 For Institutions:

- Provide counseling support
- Reduce academic pressure
- Promote mental health awareness

🚀 For Future Work:

- Build ML models for prediction
- Create dashboards for monitoring

🚀 14. FUTURE SCOPE

- Apply Machine Learning models (Logistic Regression, Random Forest)
- Build prediction system
- Develop web application/dashboard.